Raquel Forés Toribio

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Employment

2026 – present **Postdoctoral Researcher** Investigadora Doctora Júnior Prometeo
Department d'Astronomia i Astrofísica, Universitat de València, València, Spain.

2024 – 2026 **Postdoctoral Researcher** CCAPP Fellow
Department of Astronomy, The Ohio State University, OH, USA.

2020 – 2024 **Graduate grant-assisted student** Supervisor: Jose Antonio Muñoz Lozano
Departament d'Astronomia i Astrofísica, Universitat de València, València, Spain.

Education

2020 – 2024 **PhD in Physics,** Cum Laude, International mention
Universitat de València, València, Spain.

2018 – 2020 **Master's Degree in Astrophysics,** Average mark: 9.84/10
Universidad de La Laguna, Tenerife, Spain.

2014 – 2018 **Bachelor Degree in Physics,** Average mark: 9.35/10
Universitat de València, València, Spain.

Scientific Products

Journal Articles

1. **R. Forés-Toribio** and C. S. Kochanek, "The neighboring stars of N6946-BH1 and the observational characteristics of failed supernovae," *OJAp*, vol. 9, Sep. 2026. DOI: 10.33232/001c.169299.

2. J. Lillo-Box, C. Cifuentes, O. Balsalobre-Ruza, *et al.*, "CAFE follow-up of TESS hot Jupiter candidates left behind: I. Five newly confirmed planets and a false positive," *A&A*, vol. 713, A272, Sep. 2026. DOI: 10.1051/0004-6361/202660710.

3. **R. Forés-Toribio**, C. S. Kochanek, and J. A. Muñoz, "Dynamical Systematics for Time-delay Lenses and the Impact on the Hubble Constant," *ApJ*, vol. 1005, no. 2, 228, Jul. 2026. DOI: 10.3847/1538-4357/ae78c3.

4. J. García-Mejía, Z. L. de Beurs, P. Tamburo, *et al.*, "A Ground-based Transit Observation of the Long-period Extremely Low-density Planet HIP 41378 f," *AJ*, vol. 171, no. 4, 245, Apr. 2026. DOI: 10.3847/1538-3881/ae4d3d.

5. E. Mediavilla, J. Jiménez-Vicente, C. Fian, *et al.*, "Evidence for Substructure on the Inner Broad Line Region of the Lensed Quasar SDSS J1004+4112 from the Recurrence of Microlensing in the C IV Emission Line," *ApJ*, vol. 997, no. 2, 274, Feb. 2026. DOI: 10.3847/1538-4357/ae2c84.

6. B. JoHantgen, D. M. Rowan, **R. Forés-Toribio**, *et al.*, "A Systematic Search for Big Dippers in ASAS-SN," *The Open Journal of Astrophysics*, vol. 9, Feb. 2026. DOI: 10.33232/001c.156224.

7. S. W. Yee, J. N. Winn, J. D. Hartman, *et al.*, "The TESS Grand Unified Hot Jupiter Survey. III. Thirty More Giant Planets," *ApJS*, vol. 280, no. 1, 30, Sep. 2025. DOI: 10.3847/1538-4365/aded0d.

8. **R. Forés-Toribio**, B. JoHantgen, C. S. Kochanek, *et al.*, "ASASSN-24fw: An 8-month long, 4.1 mag, optically achromatic and polarized dimming event," *OJAp*, vol. 8, 114, Aug. 2025. DOI: 10.33232/001c.143105.

9 Z. Essack, D. Dragomir, P. A. Dalba, *et al.*, "Giant Outer Transiting Exoplanet Mass (GOT 'EM) Survey. VI. Confirmation of a Long-period Giant Planet Discovered with a Single TESS Transit," *AJ*, vol. 170, no. 1, 41, Jul. 2025. DOI: 10.3847/1538-3881/add88b.

10 T. R. Fairnington, J. Dong, C. X. Huang, *et al.*, "The eccentricity distribution of warm sub-Saturns in TESS," *MNRAS*, vol. 540, no. 1, Jun. 2025. DOI: 10.1093/mnras/staf759.

11 J. Ehrhardt, L. Thomas, H. Kellermann, *et al.*, "Confirmation of four hot Jupiters detected by TESS using follow-up spectroscopy from MaHPS at Wendelstein together with NEID and TRES," *A&A*, vol. 692, A220, Dec. 2024. DOI: 10.1051/0004-6361/202451404.

12 **R. Forés-Toribio**, J. A. Muñoz, C. Fian, J. Jiménez-Vicente, and E. Mediavilla, "Microlensing analysis of 14.5-year light curves in SDSS J1004+4112: Quasar accretion disk size and intracluster stellar mass fraction," *A&A*, vol. 691, A97, Nov. 2024. DOI: 10.1051/0004-6361/202347378.

13 C. Fian, J. A. Muñoz, J. Jiménez-Vicente, *et al.*, "Revealing the inner workings of the lensed quasar SDSS J1339+1310: Insights from microlensing analysis," *A&A*, vol. 689, A129, Sep. 2024. DOI: 10.1051/0004-6361/202450151.

14 C. Fian, J. Jiménez-Vicente, E. Mediavilla, *et al.*, "First Direct Evidence for Keplerian Rotation in Quasar Inner Broad-line Regions," *ApJL*, vol. 972, L7, Sep. 2024. DOI: 10.3847/2041-8213/ad654d.

15 **R. Forés-Toribio**, E. Mediavilla, J. A. Muñoz, J. Jiménez-Vicente, C. Fian, and C. del Burgo, "Single-epoch and Differential Astrometric Microlensing of Quasars," *ApJ*, vol. 970, 11, Jul. 2024. DOI: 10.3847/1538-4357/ad49a6.

16 B. J. Hord, E. M. R. Kempton, T. M. Evans-Soma, *et al.*, "Identification of the Top TESS Objects of Interest for Atmospheric Characterization of Transiting Exoplanets with JWST," *AJ*, vol. 167, 233, May 2024. DOI: 10.3847/1538-3881/ad3068.

17 C. Fian, J. A. Muñoz, **R. Forés-Toribio**, *et al.*, "Probing the structure of the lensed quasar SDSS J1004+4112 through microlensing analysis of spectroscopic data," *A&A*, vol. 682, A57, Feb. 2024. DOI: 10.1051/0004-6361/202347382.

18 F. Rescigno, G. Hébrard, A. Vanderburg, *et al.*, "A hot mini-Neptune and a temperate, highly eccentric sub-Saturn around the bright K-dwarf TOI-2134," *MNRAS*, vol. 527, pp. 5385–5407, Jan. 2024. DOI: 10.1093/mnras/stad3255.

19 S. Yoshida, S. Vissapragada, D. W. Latham, *et al.*, "TESS Spots a Super-puff: The Remarkably Low Density of TOI-1420b," *AJ*, vol. 166, 181, Nov. 2023. DOI: 10.3847/1538-3881/acf858.

20 S. W. Yee, J. N. Winn, J. D. Hartman, *et al.*, "The TESS Grand Unified Hot Jupiter Survey. II. Twenty New Giant Planets," *ApJS*, vol. 265, 1, Mar. 2023. DOI: 10.3847/1538-4365/aca286.

21 **R. Forés-Toribio**, J. A. Muñoz, C. S. Kochanek, and E. Mediavilla, "A Mass Model for the Lensing Cluster SDSS J1004+4112: Constraints from the Third Time Delay," *ApJ*, vol. 937, 35, Sep. 2022. DOI: 10.3847/1538-4357/ac8c40.

22 J. A. Muñoz, C. S. Kochanek, J. Fohlmeister, J. Wambsganss, E. Falco, and **R. Forés-Toribio**, "The Longest Delay: A 14.5 yr Campaign to Determine the Third Time Delay in the Lensing Cluster SDSS J1004+4112," *ApJ*, vol. 937, 34, Sep. 2022. DOI: 10.3847/1538-4357/ac8877.

23 S. W. Yee, J. N. Winn, J. D. Hartman, *et al.*, "The TESS Grand Unified Hot Jupiter Survey. I. Ten TESS Planets," *AJ*, vol. 164, 70, Aug. 2022. DOI: 10.3847/1538-3881/ac73ff.

24 J. Dong, C. X. Huang, G. Zhou, *et al.*, "NEID Rossiter-McLaughlin Measurement of TOI-1268b: A Young Warm Saturn Aligned with Its Cool Host Star," *ApJL*, vol. 926, L7, Feb. 2022. DOI: 10.3847/2041-8213/ac4da0.

Conferences

1. **R. Forés-Toribio**, *The failed supernova candidate N6946-BH1, its neighbors, and the defining properties of failed SNe (contributed talk), Type II Supernovae: The Chaotic Deaths of Supergiants and their Cosmic Implications, San Pedro de Atacama, Chile*, Oct. 2026.
2. **R. Forés-Toribio**, B. JoHantgen, C. S. Kochanek, *et al., ASASSN-24fw in the context of long and deep stellar dimmings in time-domain surveys (contributed talk), The 16th Gaia Science Alerts and ACME Time-Domain Workshop, Tenerife, Spain*, Sep. 2025.
3. **R. Forés-Toribio**, J. A. Muñoz, C. Fian, J. Jiménez-Vicente, and E. Mediavilla, *Quasar microlensing in galaxy clusters as a complementary approach to intracluster light (contributed talk), XVI Reunión Científica de la SEA, Granada, Spain*, Jul. 2024.
4. **R. Forés-Toribio**, J. A. Muñoz, C. Fian, J. Jiménez-Vicente, and E. Mediavilla, *Constraining the stellar mass fraction and the quasar accretion disk size of SDSS J1004+4112 with microlensing (poster), 26th International Microlensing Conference, Livermore, California, USA (online)*, Jan. 2024.
5. **R. Forés-Toribio**, J. A. Muñoz, C. Fian, J. Jiménez-Vicente, and E. Mediavilla, *Stellar mass fraction and quasar accretion disk size in SDSS J1004+4112 from photometric follow-up (contributed talk), IAU Symposium 381: Strong Gravitational Lensing in the Era of Big Data, Otranto, Italy*, Jun. 2023.
6. **R. Forés-Toribio**, *Single epoch astrometric microlensing in lensed quasars (contributed talk), 25th International Microlensing Conference, Paris, France*, Sep. 2022.
7. **R. Forés-Toribio**, J. A. Muñoz, C. S. Kochanek, and E. Mediavilla, *Remodelling the lensing cluster SDSS J1004+4112 (poster), European Astronomical Society Annual Meeting, Valencia, Spain*, Jun. 2022.

Invited Talks, Colloquiums and Seminars

1. **R. Forés-Toribio**, *The failed supernova candidate N6946-BH1, its neighbors, and the defining properties of failed SNe, CCAPP Symposium 2026, Center for Cosmology and AstroParticle Physics, The Ohio State University, Columbus, Ohio, USA*, Sep. 2026.
2. **R. Forés-Toribio**, *ASASSN-24fw in the context of long and deep stellar dimmings in time-domain surveys, CCAPP Symposium 2025, Center for Cosmology and AstroParticle Physics, The Ohio State University, Columbus, Ohio, USA*, Sep. 2025.
3. **R. Forés-Toribio**, *ASASSN-24fw: a long, deep, achromatic and polarized stellar eclipse, Ball State University Colloquium, Muncie, Indiana, USA*, Sep. 2025.
4. **R. Forés-Toribio**, *Lentes gravitatorias y su utilidad para la astronomía: El sistema B0445+123 y las observaciones con GTC, Invited talk to the GTC staff, Tenerife, Spain*, Jan. 2025.

Achievements

Grants

06/2021 – 09/2024 **Graduate grant "Atracció de Talent"**, Universitat de València.

09/2020 – 06/2021 **Graduate grant "ACIF"**, Generalitat Valenciana.

11/2017 – 07/2018 **Collaboration grant** (Undergraduate research), Ministerio de Educación y Ciencia.

Achievements (continued)

Prizes

2025 — **Premio Extraordinario de Doctorado del curso 2024/2025** *(Thesis Extraordinary Prize academic year 2024/2025).*
Awarded by Universitat de València, Valencia, Spain.

2024 — **Premi Mavi Dolç i Gastaldo 2024 a l'ús del valencià i la qualitat lingüística en les tesis doctorals** *(Mavi Dolç and Gastaldo Prize 2024 for the use of Valencian and linguistic quality in doctoral theses).*
Awarded by Universitat de València, Valencia, Spain.

2020 — **Premio Extraordinario en la Titulación de Máster Universitario en Astrofísica** *(Extraordinary Prize for the Master's Degree in Astrophysics).*
Awarded by Universidad de La Laguna, Tenerife, Spain.

— **Premio de Excelencia de Máster en la rama de conocimiento de Ciencias** *(Prize of Excellence for the Master in the branch of knowledge in Sciences).*
Awarded by Universidad de La Laguna, Tenerife, Spain.

Competitive research projects

01/09/2025 – present — **PID2024-160091NB-C32 of the Ministerio de Ciencia e Innovación**. Member of the working group.

01/09/2021 – 31/08/2025 — **PID2020-118687GB-C32 of the Ministerio de Ciencia e Innovación**. Member of the working group.

01/10/2020 – 29/06/2021 — **AYA2016-79014-C3-3-P of the Ministerio de Ciencia e Innovación**. Member of the working group.

Skills

Languages — Strong reading, writing and speaking competencies for English (Advanced Cambridge English Level), Spanish (native) and Catalan (native).

Coding — Good knowledge of Python, FORTRAN and LaTeX.
Some knowledge of Mathematica, Julia, C++ and IRAF.

Miscellaneous

International stays

08/2022 – 11/2022 — **Department of Astronomy at the Ohio State University**
with Prof. Christopher Kochanek.

Teaching experience

Course 2025-26 — **Observational Techniques in Astronomy and Photometric Observations, 15 hours**. Experto Universitario en Astrofísica Observacional (Postgraduate course), Universitat de València.

Course 2024-25 — **Observational Techniques in Astronomy and Photometric Observations, 15 hours**. Experto Universitario en Astrofísica Observacional (Postgraduate course), Universitat de València.

Course 2023-24 — **Observational Techniques in Astronomy and Photometric Observations, 15 hours**. Experto Universitario en Astrofísica Observacional (Postgraduate course), Universitat de València.

Miscellaneous (continued)

- **General Physics III (Electromagnetism and Optics), 15 hours**. Degree in Physics, Universitat de València.

Course 2022-23

- **Mechanics Laboratory, 25 hours**. Degree in Physics, Universitat de València.
- **General Physics III (Electromagnetism and Optics), 30 hours**. Degree in Physics, Universitat de València.

Course 2021-22

- **Mechanics Laboratory, 25 hours**. Degree in Physics, Universitat de València.
- **General Physics III (Electromagnetism and Optics), 30 hours**. Degree in Physics, Universitat de València.

Course 2020-21

- **Mechanics Laboratory, 50 hours**. Degree in Physics, Universitat de València.

Supervised works

Course 2023-24

- **Final Degree Project (co-supervised)**. Title: *Validación de candidatos a exoplanetas y determinación de parámetros orbitales (Exoplanet candidate validation and orbital parameters determination)* Author: Inés Pastor Espinosa. Degree in Physics, Universitat de València.

Course 2022-23

- **Final Degree Project (co-supervised)**. Title: *Seguimiento fotométrico de exoplanetas (Photometric follow-up of exoplanets)* Author: Paula Talavera Capilla. Degree in Physics, Universitat de València.

Referee experience

2026 – present

- **Referee for the MNRAS journal**.
- **Referee for the A&A journal**.

2024 – present

- **Referee for the AAS journals**.

Observing proposals

2025

- **Request for observational time at Large Binocular Telescope** as PI (observed).

2024

- **Request for observational time at Gran Telescopio Canarias** as PI (observed).
- **Request for observational time at Gran Telescopio Canarias** as co-investigator (observed).

2023

- **Request for observational time at Gran Telescopio Canarias** as PI (awarded time).

2022

- **Request for observational time at Gran Telescopio Canarias** as PI (awarded time).

2021

- **Request for observational time at Gran Telescopio Canarias** as PI (awarded time).

Extracurricular activities

2024 – present

- **CCAPP seminar coordinator**. Weekly seminar series held by the Center for Cosmology and AstroParticle Physics (CCAPP).

2022 – present

- **Member of the Spanish Astronomical Society**.

2020 – 2024

- **Member of the Department council**. Graduate students representative at the Departament d'Astronomia i Astrofísica, Universitat de València.

Miscellaneous (continued)

09/2023 — **Member of the organizing committee of E3F**. Escola Estiu Erasmus en Física (Student Summer School), Universitat de València.

Specialized training

2024 — **Responsible Conduct of Research**. Awarded by CITI program.

2023 — **Nocions essencials d'accessibilitat digital per a la docència** (Essential notions of digital accessibility for teaching). Awarded by Universitat de València.

2022 — **Julia for Scientists**. Awarded by Universitat de València.

2021 — **XXI Spanish Virtual Observatory School**. Awarded by Centro de Astrobiología.

The 5th Azarquiel School Of Astronomy. Awarded by Azarquiel School.

Introduction to Machine Learning. Awarded by Universitat de València.

Introduction to Statistics. Awarded by Universitat de València.

Time-Domain Cosmology with Strong Gravitational Lensing. Awarded by KAVLI IPMU.

2019 — **XII COEFIS** (Physics Students Congress). Awarded by Universidad de La Laguna.

2017 — **External practices at the Observatori Astronòmic de la Universitat de València**. Awarded by Universitat de València.

Outreach activities

10/2026 — **Lentes gravitatorias: las lentes de la astronomía**. Outreach talk at the Likan Antai high school, San Pedro de Atacama, Chile.

2022 – present — **Member of Astrobitos**. Blog of outreach in Spanish supported by AAS.